

Config 1 (0.05|0.9|1|1|0.65|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9017

FWER (P reject ≥ 1 | H0) : 0.0486

Max sample size (planned) : 116

ESS under H1 : 116.0

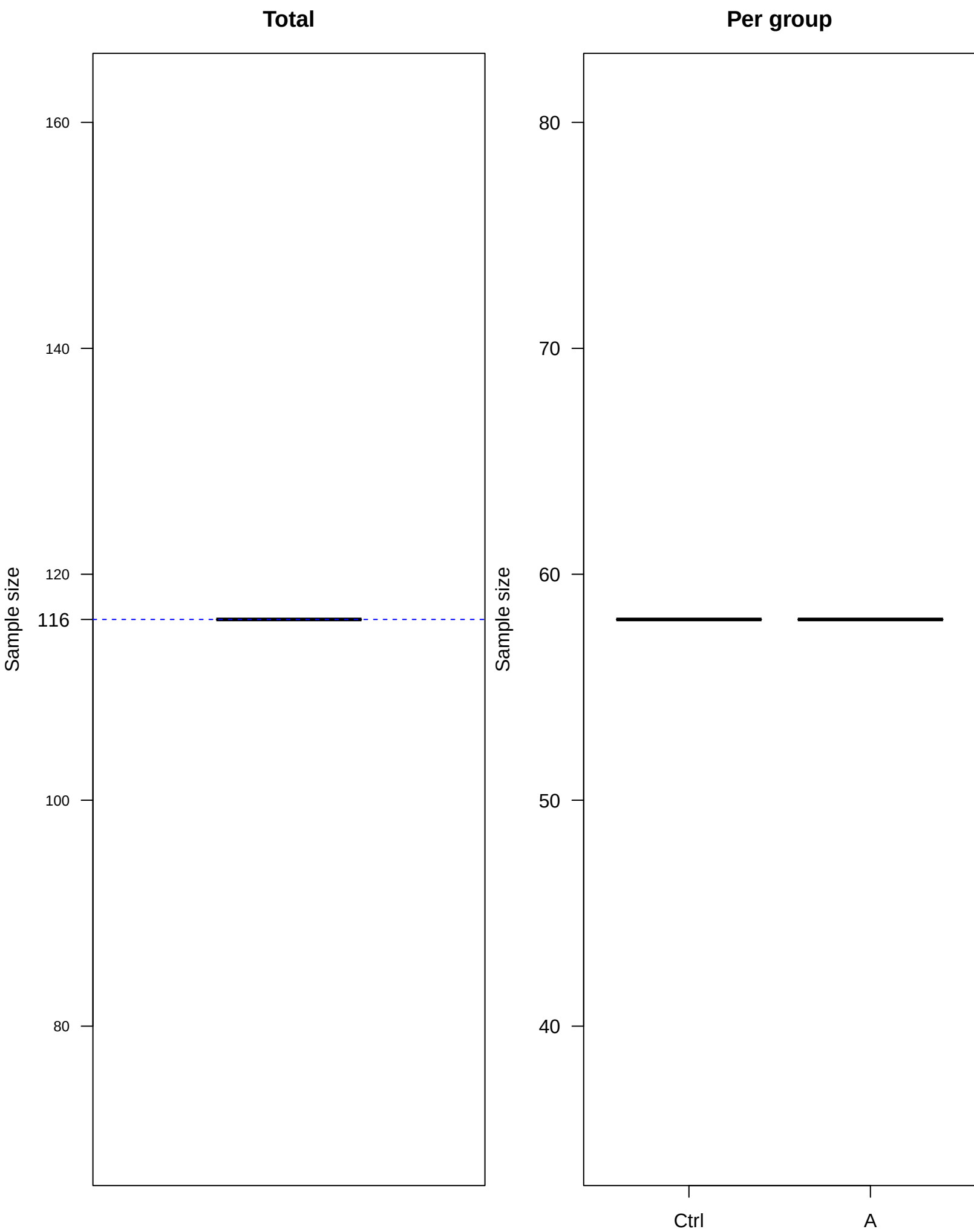
ESS under H0 : 116.0

NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

FWER is conservative vs the ../non-binding worst case.

Config 1 – sample size (H1) [non-binding, enforced]



Config 3 (0.05|0.9|1|1|0.75|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9004

FWER (P reject ≥ 1 | H0) : 0.0505

Max sample size (planned) : 38

ESS under H1 : 38.0

ESS under H0 : 38.0

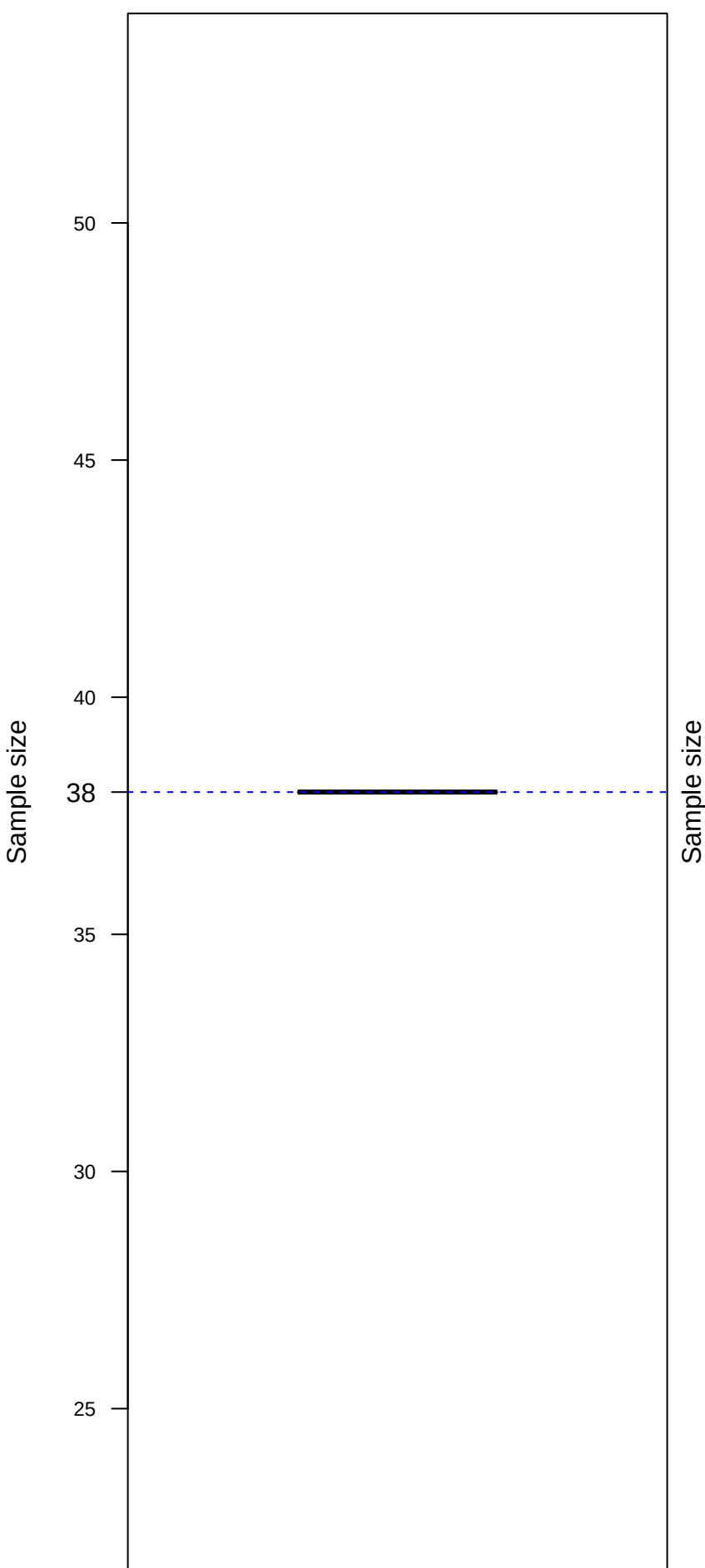
NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

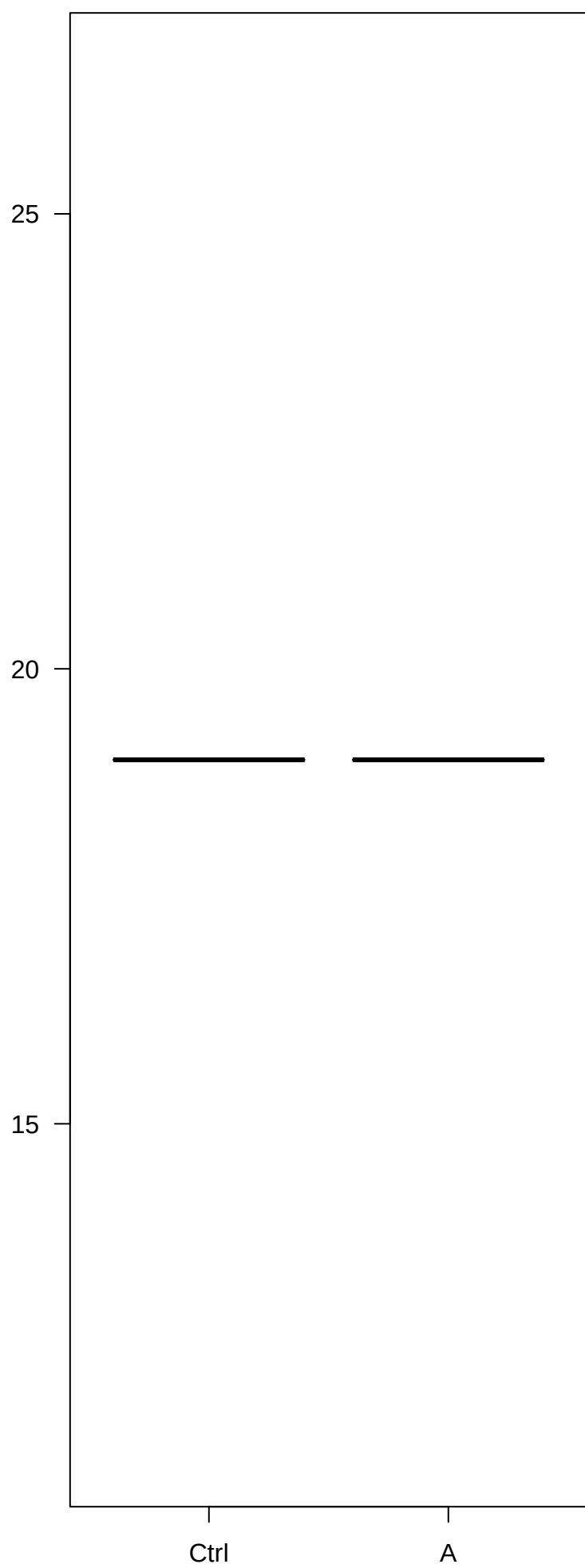
FWER is conservative vs the ../non-binding worst case.

Config 3 – sample size (H1) [non-binding, enforced]

Total



Per group



Config 5 (0.05|0.9|1|2|0.65|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9006

FWER (P reject ≥ 1 | H0) : 0.0448

Max sample size (planned) : 132

ESS under H1 : 99.7

ESS under H0 : 83.6

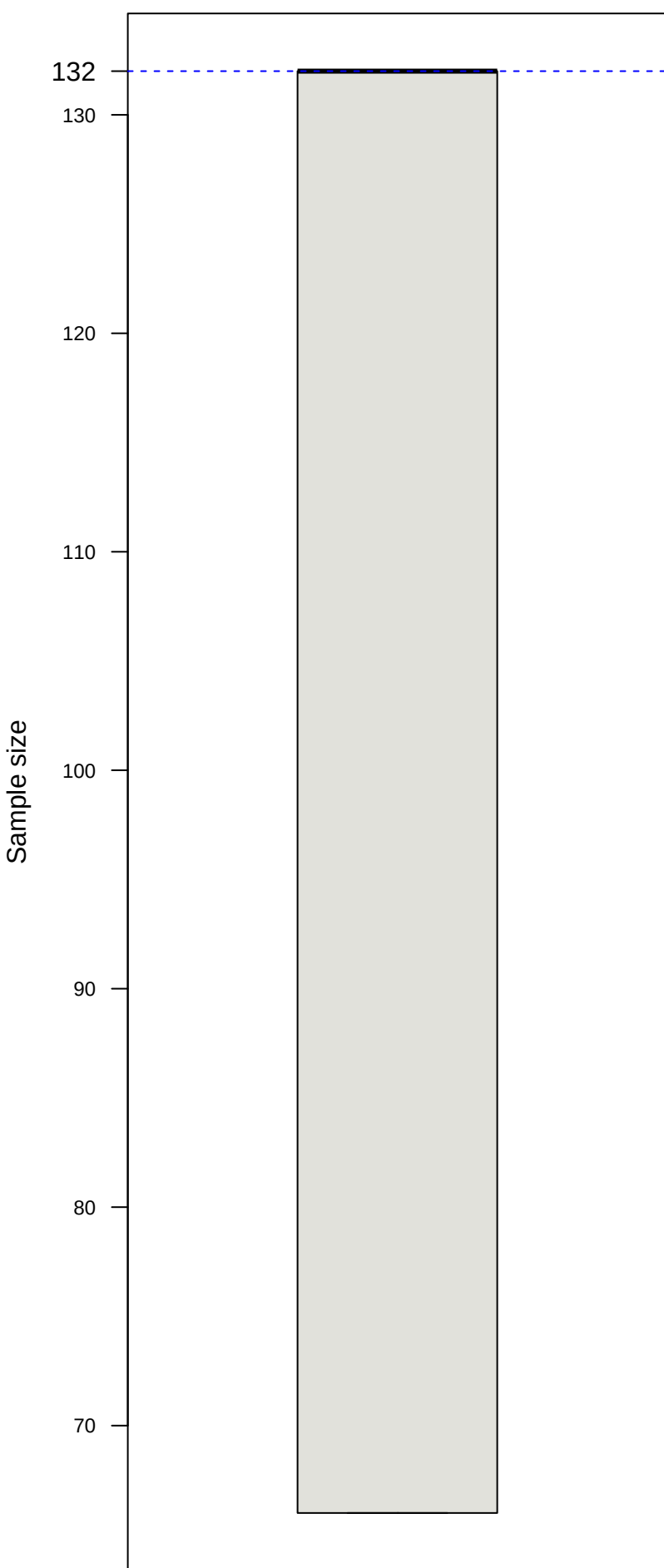
NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

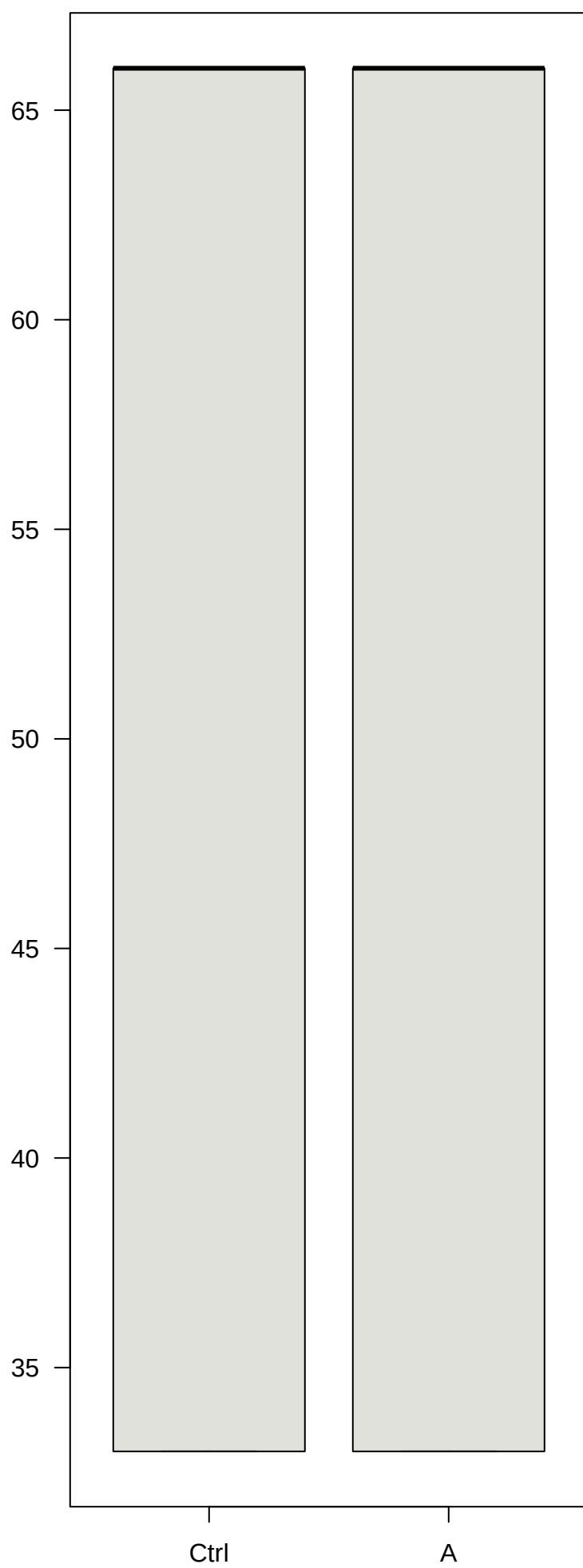
FWER is conservative vs the ../non-binding worst case.

Config 5 – sample size (H1) [non-binding, enforced]

Total



Per group



Config 7 (0.05|0.9|1|2|0.75|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9057

FWER (P reject ≥ 1 | H0) : 0.0413

Max sample size (planned) : 44

ESS under H1 : 33.1

ESS under H0 : 27.9

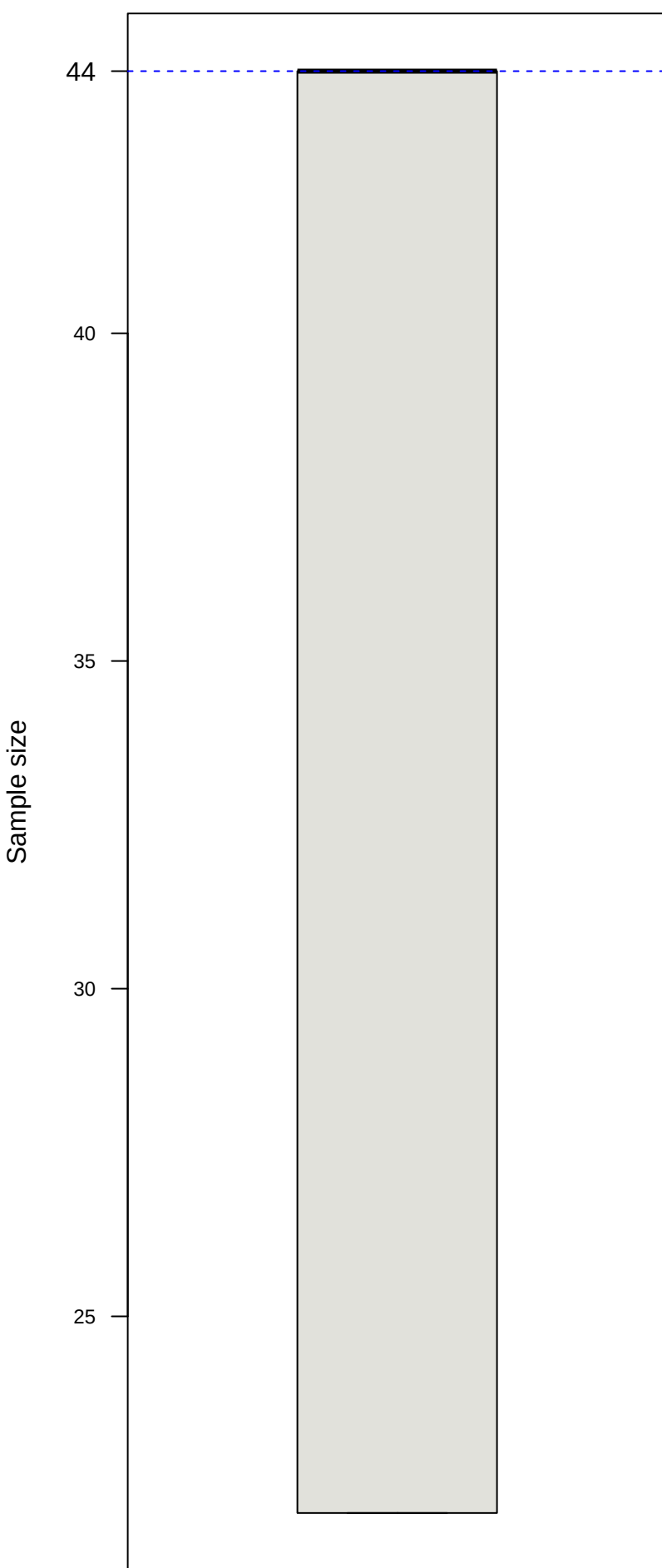
NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

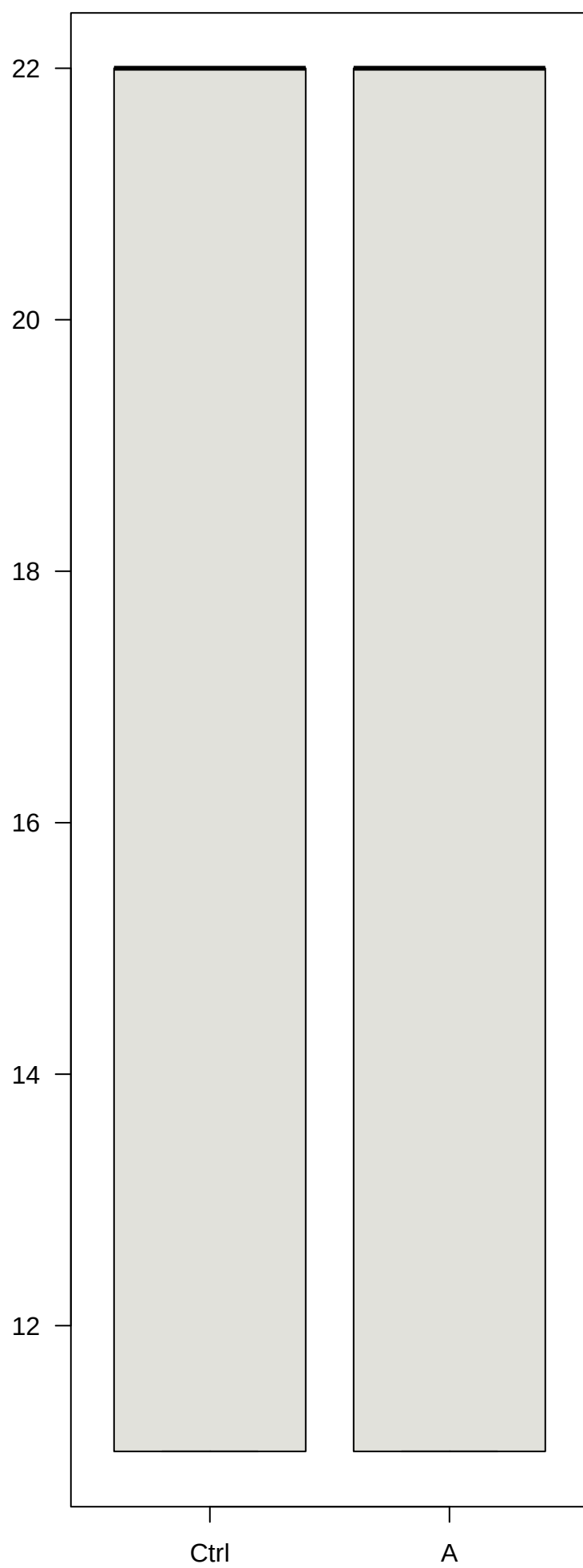
FWER is conservative vs the ../non-binding worst case.

Config 7 – sample size (H1) [non-binding, enforced]

Total



Per group



Config 9 (0.05|0.9|4|1|0.65|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9000

FWER (P reject ≥ 1 | H0) : 0.0485

Max sample size (planned) : 400

ESS under H1 : 400.0

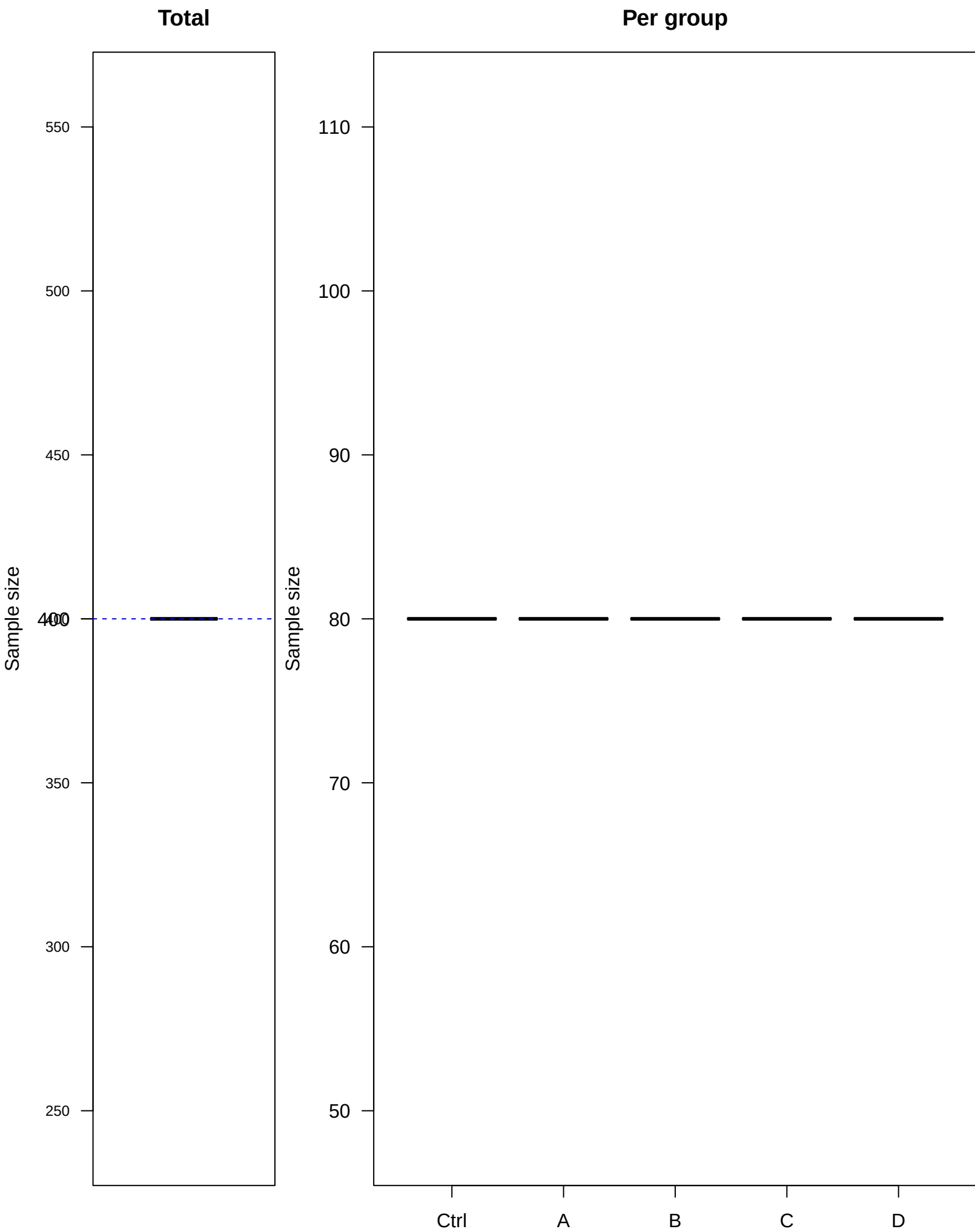
ESS under H0 : 400.0

NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

FWER is conservative vs the ../non-binding worst case.

Config 9 – sample size (H1) [non-binding, enforced]



Config 11 (0.05|0.9|4|1|0.75|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9131

FWER (P reject ≥ 1 | H0) : 0.0502

Max sample size (planned) : 135

ESS under H1 : 135.0

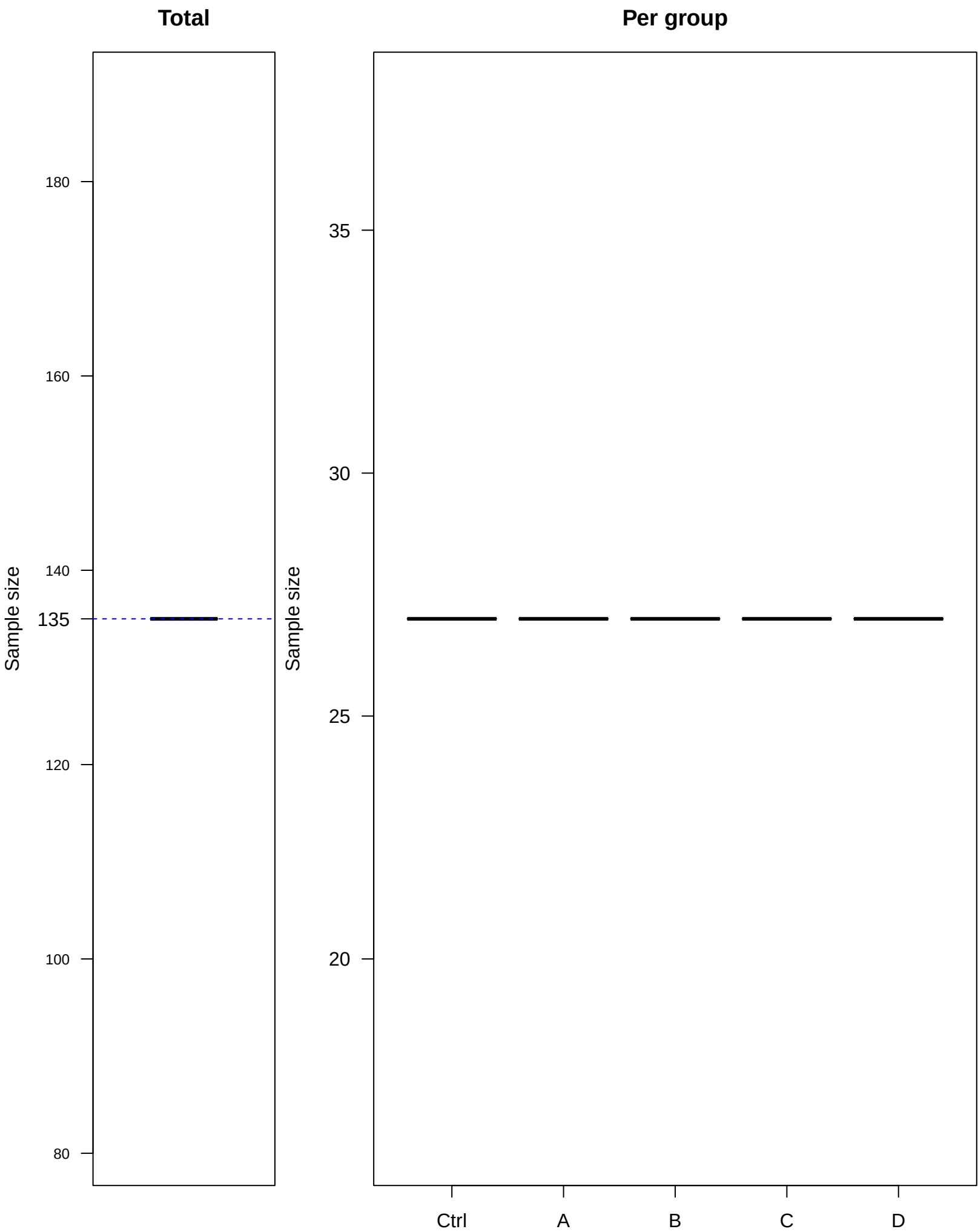
ESS under H0 : 135.0

NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

FWER is conservative vs the ../non-binding worst case.

Config 11 – sample size (H1) [non-binding, enforced]



Config 13 (0.05|0.9|4|2|0.65|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.8987

FWER (P reject ≥ 1 | H0) : 0.0460

Max sample size (planned) : 430

ESS under H1 : 285.5

ESS under H0 : 273.1

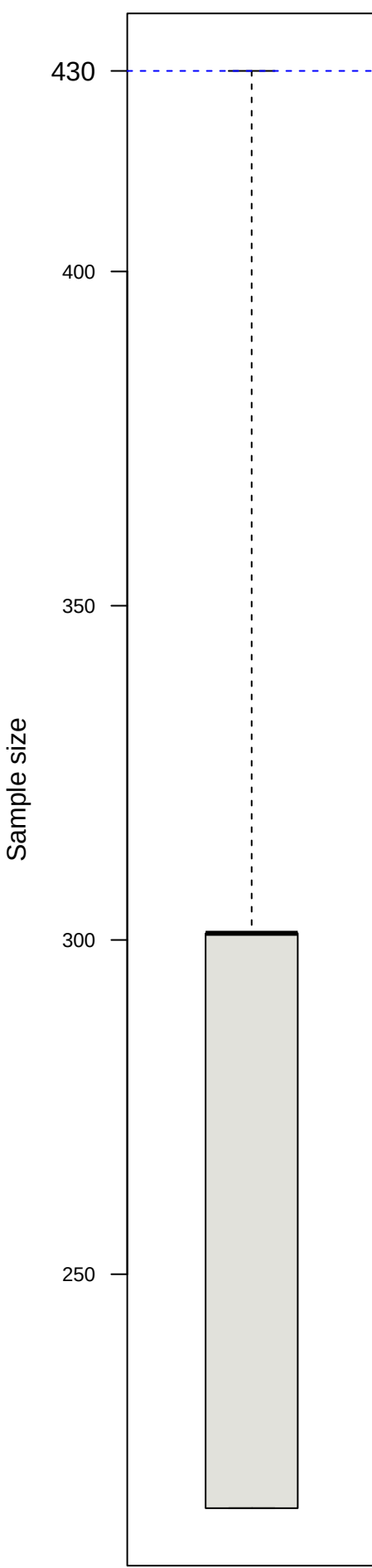
NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

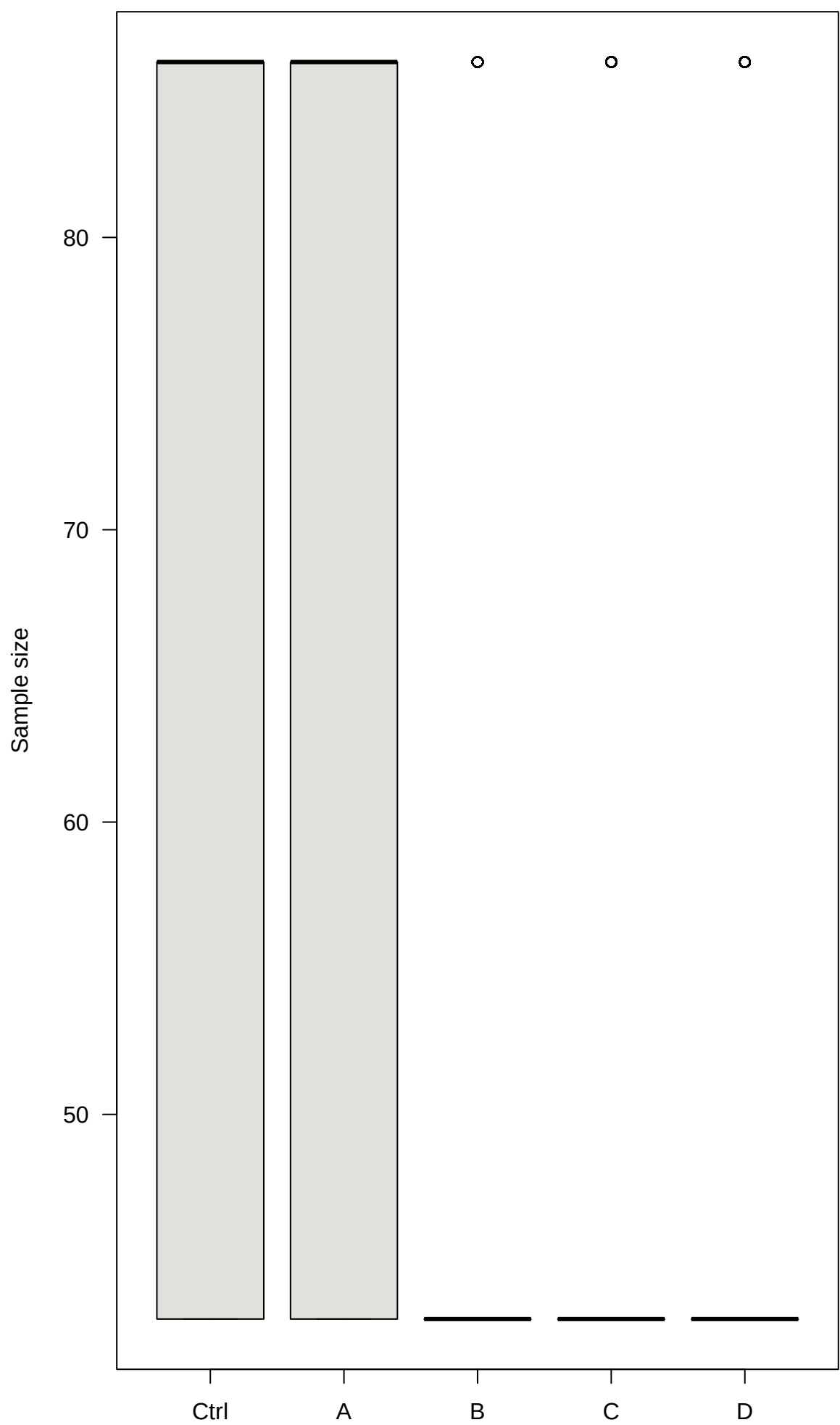
FWER is conservative vs the ../non-binding worst case.

Config 13 – sample size (H1) [non-binding, enforced]

Total



Per group



Config 15 (0.05|0.9|4|2|0.75|0.5|simultaneous) [NON-BINDING, FUTILITY ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS

Power (P reject ≥ 1 | H1) : 0.9033

FWER (P reject ≥ 1 | H0) : 0.0442

Max sample size (planned) : 140

ESS under H1 : 92.9

ESS under H0 : 88.8

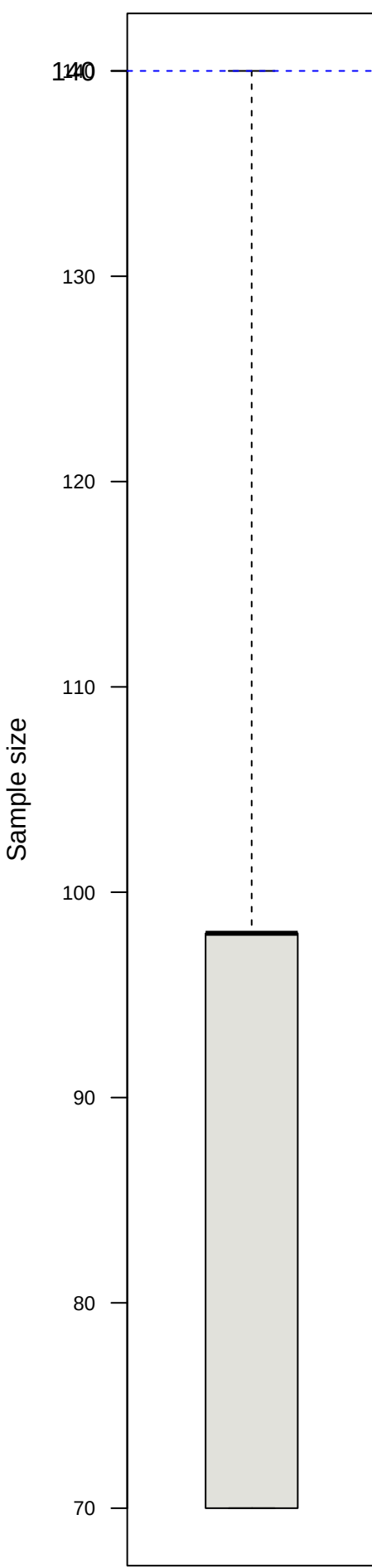
NB: non-binding boundary with futility ENFORCED.

Power and ESS are realistic (futility honoured).

FWER is conservative vs the ../non-binding worst case.

Config 15 – sample size (H1) [non-binding, enforced]

Total



Per group

